STATEMENT OF WORK

FOR

Title: Plumbing Repair, Emergency Response, and Technical Support Services

CAT: A3

REVISION: 0

Requisition No:

DATE: ***December 9, 2025***

PREPARED BY: Brian Sloat Digitally signed by Brian Sloat Date: 2025.12.18 13:54:39 -05'00'
Princeton Technical Representative: Brian Sloat

REVIEWED BY:________________________________
Facilities Engineer: Craig Shaw

APPROVED BY:________________________________
Facilities Head: Kate Morrison

PRINCETON
PLASMA PHYSICS LABORATORY
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REVISIONS

Revision	Date	Changes
0	12/09/25	Initial Issue

1.0 INTRODUCTION AND SCOPE

1.1 INTRODUCTION

The purpose of this subcontract is to provide PPPL qualified Plumbing subcontractors capable of providing on-call, repair-focused plumbing services to support PPPL's Facilities & Site Services (F&SS) department. This subcontract(s) supplements the in-house maintenance team, which performs all preventive maintenance (PM) activities. The subcontractor(s) shall have the ability to provide complete corrective maintenance and emergency response services for all major potable water and drainage systems. This scope includes, but is not limited to, 24/7/365 emergency leak response, comprehensive piping repairs, advanced diagnostic testing, line camera inspections, and hydro-mechanical drain clearing across all system types. PPPL's portfolio includes a mix of plumbing systems including sanitary, storm, domestic, laboratory, gas, and specialty plumbing systems.

PPPL will issue individual Job Orders for specific scopes of work with associated costs. Each Job Order will detail the exact nature of the required repair or troubleshooting, the specific plumbing system involved, the expected timeline for completion, and the agreed-upon cost for the services outlined. Subcontractors will execute work only upon receiving an authorized Job Order.

1.2 SCOPE

The Contractor shall furnish all labor, materials, equipment, lifts, tools, and fall-protection required to perform plumbing repairs and diagnostic services. All work shall comply with DOE directives, OSHA regulations, NFPA, IBC, Princeton Plasma Physics Laboratory (PPPL) and environmental regulations. Scope of services include, but are not limited to:

1.3 TASKS

1.3.1 Domestic Water Systems

1.3.1.1 Leak detection and repair of domestic water piping

1.3.1.2 Repair/replacement of valves, PRVs, mixing valves, and backflow preventers

1.3.1.3 Water heater repairs or replacements

1.3.1.4 Fixture repairs (sinks, faucets, showers, emergency fixtures)

1.3.1.5 Recirculation pump repairs

1.3.1.6 Emergency isolation and system restoration

1.3.2 Sanitary & Sewer Systems

1.3.2.1 Clearing drain and sewer blockages using mechanical or jetting methods

1.3.2.2 Repairs or replacement of sanitary piping, traps, and branch lines

1.3.2.3 Sewer odor troubleshooting

1.3.2.4 Camera inspections and root-cause analysis

1.3.3 Stormwater Systems

1.3.3.1 Repairs to roof drains, conductors, and storm piping

1.3.3.2 Clearing storm line blockages

1.3.3.3 Sump pump and ejector system repairs

1.3.4 Laboratory & Specialty Plumbing Systems

1.3.4.1 DI/RO water system repairs (non-PM)

1.3.4.2 Chemical waste piping repairs (where permitted)

1.3.4.3 Lab sink and emergency wash fixture repairs

1.3.4.4 Troubleshooting pressure or quality issues in process water systems

1.3.5 Gas Piping Systems

1.3.5.1 Repair or replacement of natural gas, lab gas, and process gas piping

1.3.5.2 Leak identification and correction.

1.3.5.3 Valve replacements and pressure troubleshooting

1.3.6 Mechanical Support

1.3.6.1 Pump repairs and replacements (booster, sump, ejector)

1.3.6.2 Heat exchanger plumbing repairs

1.3.6.3 Support for tie-ins and emergency mechanical replacements

1.3.7 Emergency Services

1.3.7.1 Burst pipes, major leaks, sewer backups

1.3.7.2 Domestic water system failures

1.3.7.3 Flooding related to plumbing infrastructure

1.3.7.4 Plumbing failures affecting mission-critical systems

1.3.7.5 Expected response times: 1–2 hours for critical events; 4–24 hours for non-urgent events

1.3.8 Troubleshooting & Technical Support

1.3.8.1 Root-cause analysis of persistent failures

1.3.8.2 Pressure and flow testing

1.3.8.3 Camera inspections with documentation

1.3.8.4 Leak detection with ultrasonic/thermal tools

1.3.8.5 Water quality troubleshooting related to infrastructure

1.3.8.6 Code compliance recommendations

1.3.9 Replacement & Installation Services

1.3.9.1 Installation of piping, fixtures, pumps, traps, regulators, and meters

1.3.9.2 Small-scope capital repairs as directed

1.3.9.3 Support for capital projects under separate agreements

1.4 EXCLUSIONS

1.4.1 This contract does not include large capital construction, engineering design services outside troubleshooting, HVAC-only repairs unrelated to plumbing, welding beyond plumbing code scope unless specified in the Job Order.

2.0 APPLICABLE DRAWINGS & SUPPORTING DOCUMENTS

PPPL PTR will provide any documents needed for the Subcontractor to perform the individual Job Order.

3.0 RESPONSIBILITIES

3.1 PRINCETON PLASMA PHYSICS LABORATORY

3.1.1 PPPL will identify a Procurement Representative and a Princeton Technical Representative (PTR) for this Subcontract.

3.1.2 All communications on administrative matters shall be directed to the PPPL Procurement Representative.

3.1.3 All communications on technical matters shall be directed to the assigned PTR.

3.1.4 All work activities (scheduling, de-energizing, equipment staging and deliveries, and site access authorization) must be coordinated with the PTR to minimize systems down time and shall not have any negative impact on experimental operations.

3.2 SUBCONTRACTOR

3.2.1 General Requirements

3.2.1.1 The Subcontractor shall provide a point of contact for each individual Job Order between PPPL and the Subcontractor. The Subcontractor shall notify the PPPL PTR of any site visit at least 24 hours prior to arrival.

3.2.1.2 The Subcontractor shall provide all labor, materials, tools, equipment, and supervision services necessary to complete each individual Job Order in accordance with this Statement of Work and individual Job Order scope of work. Means and methods to complete individual Job Order are the responsibility of the Subcontractor and shall be reviewed by PPPL.

3.2.1.3 Subcontractor personnel expected to be onsite for more than 40 hours in a year shall be required to receive General Employee Training (GET) through the Princeton University LMS System and will be badged for PPPL access.

3.2.1.4 All subcontractors and lower-tier subcontractors who are not badged must provide the PTR their names, citizen status and contact information 48 hours prior to arrival. Access to the property will only be granted to those subcontractors and lower-tier subcontractors who present a REAL ID–compliant ID or a valid passport at the gate

3.2.1.5 All work should be completed during the normal working hours of 7:00 AM to 3:30 PM, Monday through Friday. Off hours work shall be communicated and approved by PPPL prior to the start of work. Noise restrictions may be placed on the contractor during the performance of off hours work. See Attachment 1 - Notification Requirements Off-Normal Events and Issues

3.2.1.6 The Subcontractor shall notify PPPL for approval of any lower-tier subcontractors prior to their arrival onsite.

3.2.1.7 The Subcontractor shall protect any and all infrastructure, equipment and property within and en route to the individual Job Order work area to prevent damage work activities. The Subcontractor shall maintain a clean work area at all times. All trash is to be removed daily. Any loose materials, tools and equipment are to be secured daily.

3.2.1.8 The Subcontractor shall provide supervision for traffic maintenance as necessary to minimize interference with normal use of parking lots, roads/driveways, sidewalks, loading docks, exterior doors, and adjacent facilities during construction operations.

3.2.1.9 Pre-approval is required before special purpose vehicles or equipment such as mobile cranes, forklifts, man-lifts, etc. can be delivered and accepted onsite. The

Subcontractor must submit to PPPL operator training qualifications, as well as, equipment maintenance and inspection records for all special purpose equipment. At a minimum, signed and dated annual inspection records (check-list type) shall include model and serial number for each specific piece of equipment. In addition, the equipment manual shall be submitted for review and provided on site along with the equipment. Upon delivery and prior to use, PPPL reserves the right to perform equipment inspections to verify that the equipment is acceptable for usage onsite. For additional requirements not mentioned, refer to PPPL Engineering Standards ES-MECH-007, ES-MECH-010, ES-MECH-011, ES-MECH-012.

3.2.1.10 For all hoisting and rigging, a Lift Plan shall be submitted prior to all activity. Multiple lift plans may be required depending on changes of conditions, loads or equipment used. The lift plan shall include:

3.2.1.10.1 Product data for all applicable hoisting & rigging equipment (with serial numbers for major components)

3.2.1.10.2 Periodic (minimum annual) equipment inspection reports (with serial numbers) for all equipment

3.2.1.10.3 Monthly hook and wire rope inspection reports

3.2.1.10.4 Lift data sheet detailing loads, center of gravity, attachment points, component capacities, calculations, lift angle, and swing location. Requires signature by the Subcontractor's Lift Engineer.

3.2.1.10.5 Qualifications of the Rigger and Signal Person. If qualifications are not available, an officer from the Subcontractor's firm shall sign that the stated Rigger and Signal Person are qualified per OSHA 1926.1400.

3.2.1.10.6 Equipment manual and load charts.

4.0 QUALIFICATIONS

4.1 Minimum 5 years' experience with commercial and/or institutional plumbing systems

4.2 Ensure all technicians are qualified and trained for the type of work requested in the Job Order

4.3 Demonstrated experience with plumbing system types (all subcategories listed under 1.3 Tasks)

4.4 Provide 3 customer references

5.0 ENVIRONMENT, SAFETY, AND HEALTH

All onsite work shall comply with the PPPL MNL-007 Subcontractor Requirements Manual (September 2025)

6.0 ATTACHMENTS

12.1 Attachment #1 – Notification Requirements for Off-Normal Events and Issues

12.2 Attachment #2 – DOE Headmark List

7.0 DOCUMENTATION & DELIVERABLES

#	Physical Deliverables Required	When Deliverable Is Required
1	Completed Repair for each individual Job Order	Job Order Completion
Exceptions (Add justification for any missing physical deliverables that will not be received):		

#	Document Deliverables Required	When Deliverable Is Required	Deliverable format (paper, electronic etc.)
2	All Service Reports ● Material list and repair descriptions ● Before/After photos	Within 48 hrs	Electronic/Paper
3	Testing documentation (per individual Job Order)	At completion	Electronic
4	Cost Proposals for repairs and/or replacements	As requested	Electronic
Exceptions (Add justification for any missing document deliverables that will not be received):			

Princeton Technical Representative/COG:____

Attachment #1

Notification Requirements Off-Normal Events and Issues

Note: Contacting the PPPL Emergency Services Unit at X3333 to obtain emergency assistance and taking immediate actions to protect workers, guests and visitors takes precedence over the notification requirements described herein.

The Subcontractor shall notify PPPL within 15 minutes of becoming aware of off-normal events and issues; off-normal events and issues include:

- Personnel injuries of any kind
- Near misses that could have resulted in significant (recordable injury or worse) worker injuries
- Uncontrolled or unforeseen personnel exposure to chemical, biological or physical hazards (e.g., noise, laser, ultraviolet light, heat, etc.)
- Fire emergency of any kind
- Any unexpected discovery of an uncontrolled hazardous energy source (e.g., live electrical power circuit, etc.) not including discoveries made by zero-energy checks and other precautionary investigations made before work is authorized to begin
- Any failure to follow a prescribed hazardous energy control process (e.g., lockout/tagout)
- Damage of any kind to Laboratory property
- Damage of any kind to personal property owned by PPPL employees, guests and visitors
- Any spills of chemicals, fuels, lubricants, etc.
- Other off-normal events and issues deemed by the Subcontractor to require notification

In the event of an off-normal event or issue, the Subcontractor and its lower-tier subcontractors SHALL:

- Notify PPPL Emergency Services Unit at X3333 for emergency response, if required
- Take immediate action to evacuate and post affected area, if required
- Cease related work or work in affected area
- Take actions to post and safe-off affected area
- Within 15 minutes: Contact Subcontractor Site Supervisor, PPPL PTR, ESH Construction Safety Engineer and ESH Head

Regarding these off-normal events or issues, no recovery actions (other than those immediately required to preserve life safety) or repairs are to take place until they are reviewed by the Subcontractor and PPPL, and specifically authorized by the PPPL PTR or ESH Department Head. Resumption of work that has been stopped by PPPL due to off-normal events and issues must follow the provisions of PPPL Policy P-012.

Attachment #2

DOE Headmark List

ANY HOLT ON THIS LI.ST SHOULD BE TREATED AS DEFECTIVE W.ITUOUT FURTI-nm. TESTING.

ALL GRADE 5 ANO GRADE 8 FASTENERS OF FOREIGN ORIGIN WHICH DO NOT BEAR ANY MANUFACTURERS' HE!ADMARKS:

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GRADE 5 FASTENERS WITH THE FOLLOWING MANUFACTURERS' HEA DMARKS:

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GRADE 8 FA!STENERS WITH THE FOLLOWING MANUFACTURES' HEADMAAKS:

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©	A	Asahi Mfr, (JP)	®	KS	Kosaka Kogyo (JP)
®	NF	Nippon Fasleners (JP)		RT	Takai Ltd (JP)
				FM	Fastener Co of Japan (JP)
©	H	HInomoto Metal (JP)	®		
©	M	MlnamJc:{a Sloybo (JP)	®	KV	Kyoel Mfg {JP)
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GRADE 8.2 FASTENERS WITH THE FOLLOWING HEADMARKS:

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©	KS	KO$aktl Kogyo (JP)

GRADE A325 FASTENERS{BENNETT DENVER TARGET ONLY) WITH THE FOLLOWING HEA.DMARKS:

		MARK	MANUFACTURER
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Type,3	®		

Key: CA-C11�da, JP-Japan, TW-Talwan, YU-Yugoelavla

Attachment 1 - Deliverables List

#	Physical Deliverables Required	When Deliverable Is Required
1		
Exceptions (Add justification for any missing physical deliverables that will not be received):		

#	Document Deliverables Required	When Deliverable Is Required	Deliverable format (paper, electronic etc.)
2			
Exceptions (Add justification for any missing document deliverables that will not be received):			

Princeton Technical Representative/COG:____